Reference: Arakawa's lectures, 1605.00138.

Idea behind slices.

Example

$X=\mathrm{Mat}_{n\times n}(\mathbb{C})$, $G=GL_n(\mathbb{C})$.

$G\curvearrowright X$ by conjugations.

$$S=\left\{\begin{pmatrix}\lambda_1 & & 0\\ & \ddots & \\ 0 & & \lambda_n\end{pmatrix}\middle|\,\lambda_i\neq\lambda_j,\ i\neq j\right\}\cong\mathbb{C}^n.$$

Then the restriction

$$\mathbb{C}[X]^G\longrightarrow\mathbb{C}[S]\cong\mathbb{C}[\lambda_1,\dots,\lambda_n].$$

is injective.

It maps to $\mathbb{C}[\lambda_1,\dots,\lambda_n]^{S_n}$ onto

- Actually it is surjective (as $\mathbb{C}[\lambda_1,\dots,\lambda_n]^{S_n}\cong\mathbb{C}[e_1,\dots,e_n]$).

§ Companion matrices and invariant polynomials.

Set $G=GL_n$, $\mathfrak{g}=\mathfrak{gl}_n$.

Consider

$$\mathcal{C}=\left\{\underbrace{\begin{pmatrix}0 & 0 & \cdots & 0 & -a_1\\ 1 & 0 & \cdots & 0 & \vdots\\ & \ddots & \ddots & 0 & \vdots\\ 0 & & \cdots & 1 & -a_n\end{pmatrix}}_{\alpha}\middle|\,a_i\in\mathbb{C}\right\}$$ – Companion matrices.

Properties:

- $\det(\lambda-\alpha)=a_1+a_2\lambda+\dots+a_n\lambda^{n-1}+\lambda^n$.
- $G.\mathcal{C}=\{\text{cyclic operators in }\mathrm{Mat}_{n\times n}\}$.
- $G.\mathcal{C}$ is open dense in $\mathrm{Mat}_{n\times n}(\mathbb{C})$.
- Any G-orbit intersects $\mathcal{C}$ in at most one point.

Theorem

The restriction map

$$R:\mathbb{C}[\mathfrak{g}]^G\longrightarrow\mathbb{C}[\mathcal{C}].$$

Is an isomorphism.

Proof.

$\forall f\in\mathbb{C}[\mathfrak{g}]^G$ if $f|_{\mathcal{C}}=0\Rightarrow f(A)=0\ \forall$ cyclic operator $\Rightarrow$ $\Rightarrow f=0.\Rightarrow R$ is injective.

$\forall\tilde{f}\in\mathbb{C}[\mathcal{C}]$, define $f(x)=\tilde{f}(c_1(x),\dots,c_n(x))$, where $e_i(x)$ is defined by $\det(\lambda-x)=\sum_{i=1}^{n-1}c_i(x)\lambda^i+\lambda^n$.

Then clearly $f|_{\mathcal{C}}=\tilde{f}$ as $c_i\left(\begin{pmatrix}0 & & -a_1\\ 1 & 0 & \vdots\\ 0 & 1 & -a_n\end{pmatrix}\right)=a_i$.

$\blacksquare$

Note that

$$\mathcal{C}=f+\mathfrak{a},\text{ where } f=\begin{pmatrix}1 & & 0\\ & \ddots & \\ 0 & & 1\ \ 0\end{pmatrix};\ \mathfrak{a}=\left\{\begin{pmatrix}0 & \begin{matrix}*\\ \vdots\\ *\end{matrix}\end{pmatrix}\right\}.$$

Fix standard $\mathfrak{n},\mathfrak{b}$ and pairing by $\mathrm{Tr}(\cdot\,,\cdot)$ on $\mathfrak{g}$.

Define

$$\begin{aligned}\chi:\mathfrak{n}&\longrightarrow\mathbb{C}\\ x&\longmapsto(f,x)=\mathrm{Tr}(f.x).\end{aligned}$$

Note: $\chi([\mathfrak{n},\mathfrak{n}])=0$.

- $\mathcal{O}_N(\chi)=\{\chi\}$.
- Let $\mu:\mathfrak{g}^*\longrightarrow\mathfrak{n}^*$ be restriction map.

  Then $\mu^{-1}(\chi)=\chi+\mathfrak{n}^\perp\cong f+\mathfrak{b}$ is N-invariant set.

Theorem (Kostant)

The action

$$\begin{aligned}N\times\mathcal{C}&\xrightarrow{\varphi}f+\mathfrak{b}\\ (g,x)&\longmapsto g.x.g^{-1}\end{aligned}$$

Is an N-equivariant isomorphism (of affine varieties)

Proof.

$[\mathfrak{n},f+\mathfrak{b}]\subset\mathfrak{b}\Rightarrow N.(f+\mathfrak{b})\subset f+\mathfrak{b}$.

- Any matrix in $f+\mathfrak{b}$ is cyclic hence φ is surjective.
- Two companion matrices are conjugated iff they are equal.

Stabilizer of a companion matrix in N is trivial:

$\forall\xi\in\mathfrak{n}$ decompose ξ w/ $\mathfrak{sl}_2$ triple (e,h,f) weights.

Then $\xi=\sum_{i>0}\xi_i$ let i_0 be minimal s.t. $\xi_{i_0}\neq0$.

Then $[f+\alpha,\xi]=\underset{\neq0}{[f,\xi_{i_0}]}+\text{higher weights}$.

Hence $Z_{\mathfrak{n}}(f+\alpha)=\{0\}\Rightarrow Z_N(f+\alpha)$ is finite $\Rightarrow$ trivial.

Then if $\varphi((g,x))=\varphi((\tilde{g},\tilde{x}))\Rightarrow$ 1) $x=\tilde{x}$
2) $g.\tilde{g}^{-1}\in Z_N(x)\Rightarrow g=\tilde{g}$.

Hence φ is injective.

We conclude that φ is bijection.

- Then φ is birational (some alg. geometry)
- Since $f+\mathfrak{b}$ is normal, φ^{-1} is regular (Zarissky main theorem) $\blacksquare$

Corollary.

The restriction maps

$$\mathbb{C}[\mathfrak{g}]^G\xrightarrow[\varphi_1]{}\mathbb{C}[f+\mathfrak{b}]^N\xrightarrow[\varphi_2]{}\mathbb{C}[\mathcal{C}]$$

Are isomorphisms.

Proof. $\varphi=\varphi_2\circ\varphi_1$ is an isomorphism $\Rightarrow\varphi_2$ is surjective, φ_1 is inj.

$\forall f\in\mathbb{C}[f+\mathfrak{b}]^N\Rightarrow f|_{\mathcal{C}}=0\Rightarrow f|_{f+\mathfrak{b}}=0$ as $f+\mathfrak{b}=N.\mathcal{C}\Rightarrow\varphi_2$ is inj.

$\forall f\in\mathbb{C}[f+\mathfrak{b}]^N\Rightarrow f=\varphi_1(\varphi^{-1}(\varphi_2(f)))$ $\blacksquare$

§ Transversality.

Lemma The space $f+\mathfrak{b}$ intersect $\mathcal{O}_G(f)$ at f transversely.

Proof:

Due to $\mathfrak{sl}_2$-decomposition.

$T_f(f+\mathfrak{b})=\mathfrak{b}$, $T_f(\mathcal{O}_G)=[f,\mathfrak{g}]\supset\mathfrak{n}_-\Rightarrow T_f(f+\mathfrak{b})+T_f(\mathcal{O}_G)=\mathfrak{g}$

$\blacksquare$

Def.

Let (e,h,f) be an $\mathfrak{sl}_2$-tripple corresp. to f.

This defines embedding $\tilde{\gamma}:SL_2(\mathbb{C})\hookrightarrow G$.

Then $\gamma:\mathbb{C}^\times\longrightarrow G$ is defined as $\tilde{\gamma}|_{\mathbb{C}^\times=\left\{\begin{pmatrix}t & 0\\ 0 & t^{-1}\end{pmatrix}\middle|\,t\in\mathbb{C}^\times\right\}\subset SL_2}$.

Define

$$\begin{aligned}\rho:\mathbb{C}^\times\times\mathfrak{g}&\longrightarrow\mathfrak{g}\\ (t,x)&\longmapsto t^2\mathrm{Ad}_{\gamma(t)}(x).\end{aligned}$$

Exercise ρ contracts $f+\mathfrak{b}$ to f. Namely

$\forall x\in f+\mathfrak{b}$, $\lim_{t\to0}(\rho(t,x))=f$.

Prop. The affine space $f+\mathfrak{b}$ intersects adjoint G-orbits transversely at any point of $f+\mathfrak{b}$.

Proof:

This is true at $f\Rightarrow$ holds in a (usual top.) neighborhood of f.

But $f+\mathfrak{b}$ is cotracted to f by torus action and this action preserves transversality. $\blacksquare$

§ $\mathcal{S}=\mathcal{C}$ as a reduction.

Recall: $\mathfrak{g}^*$ is a Poisson variety w/ KKS bracket

$\{x,y\}=[x,y]\ \forall x,y\in\mathfrak{g}=(\mathfrak{g}^*)^*$.

Lemma:

- The coadjoint action $G\curvearrowright\mathfrak{g}^*$ is hamiltonian w/ moment map $\mu=\mathrm{Id}:\mathfrak{g}^*\longrightarrow\mathfrak{g}^*$.
- Let $G\curvearrowright X$ be a hamiltonian action on X w/ $\mu^G:X\to\mathfrak{g}^*$. Let $H\leq G$, then restriction $H\curvearrowright X$ is also hamiltonian w/ $\mu^H=r^{\mathfrak{g}}_{\mathfrak{h}}\circ\mu^G$, $r^{\mathfrak{g}}_{\mathfrak{h}}:\mathfrak{g}^*\longrightarrow\mathfrak{h}^*$-restriction, $\mathfrak{h}=\mathrm{Lie}(H)$.

$\blacksquare$ (Exercise)

Consider adjoint action of $N\curvearrowright\mathfrak{g}^*$.

Then it is hamiltonian w/ moment map given by restriction $r=\mu:\mathfrak{g}^*\longrightarrow\mathfrak{n}^*$.

As we've seen:

$$\mathcal{O}_N(\chi)=\{\chi\}$$

Then $\mathfrak{g}/\!\!/_{\mathcal{O}_N(\chi)}N=\mu^{-1}(\mathcal{O}_N(\chi))/N=\mu^{-1}(\chi)/N=(f+\mathfrak{b})/N=$

$=(\mathcal{S}\times N)/N\cong\mathcal{S}$.

Cor. $\mathcal{S}$ is a Poisson variety.

Q: What is the bracket?

Recall: We have

$\mathbb{C}[X/\!\!/_0G]=\mathbb{C}[\mu^{-1}(0)]^G$, Poisson bracket induced from by restricting from X.

Hence $\mathbb{C}[\mathfrak{g}/\!\!/_\chi N]=\mathbb{C}[\chi+\mathfrak{n}^\perp]^N=\mathbb{C}[f+\mathfrak{b}]^N\cong\mathbb{C}[\mathfrak{g}]^G$.

But $\forall f_1,f_2\in\mathbb{C}[\mathfrak{g}]^G$, $\{f_1,f_2\}=0\Rightarrow$ We get 0 bracket.
Why?

- Move towards generalization.

DrawNote